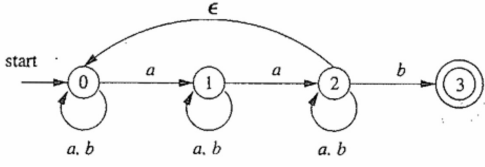


Ex 3.7.1

2) 图 3-29

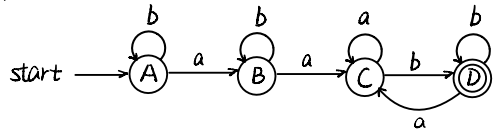
NFA:



$A = \epsilon\text{-closure}(0) = \{0\}$
 $Dtrans[A, a] = \{0, 1\} =: B$
 $Dtrans[A, b] = \{0\} = A$
 $Dtrans[B, a] = \{0, 1, 2\} =: C$
 $Dtrans[B, b] = \{0, 1\} = B$
 $Dtrans[C, a] = \{0, 1, 2\} = C$
 $Dtrans[C, b] = \{0, 1, 2, 3\} =: D$
 $Dtrans[D, a] = \{0, 1, 2\} = C$
 $Dtrans[D, b] = \{0, 1, 2, 3\} = D$

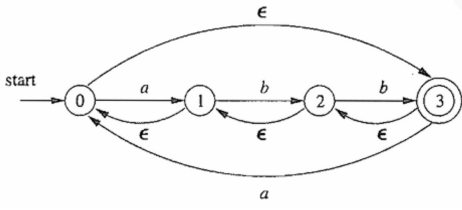
NFA states	DFA states	a	b
$\{0\}$	A	B	A
$\{0, 1\}$	B	C	B
$\{0, 1, 2\}$	C	C	D
$\{0, 1, 2, 3\}$	D	C	D

DFA:



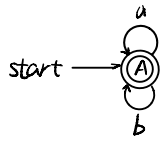
3) 图 3-30

NFA:



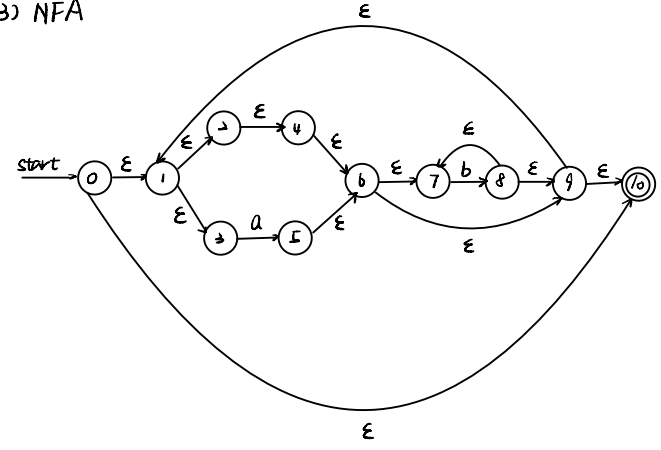
$A = \epsilon\text{-closure}(0) = \{0, 1, 2, 3\}$
 $Dtran[A, a] = \{0, 1, 2, 3\} = A$
 $Dtran[A, b] = \{0, 1, 2, 3\} = A$

DFA:



Ex 3.7.3

3) NFA



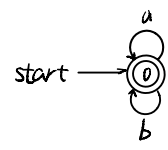
$A = \epsilon\text{-closure}(0) = \{0, 1, 2, 3, 4, 6, 7, 9, 10\}$
 $Dtrans[A, a] = \{1, 2, 3, 4, 5, 6, 7, 9, 10\} =: B$
 $Dtrans[A, b] = \{1, 2, 3, 4, 6, 7, 8, 9, 10\} =: C$
 $Dtrans[B, a] = B$ $Dtrans[C, a] = B$
 $Dtrans[B, b] = C$ $Dtrans[C, b] = C$

NFA states	DFA states	a	b
$\{0, 1, 2, 3, 4, 6, 7, 9, 10\}$	A	B	C
$\{1, 2, 3, 4, 5, 6, 7, 9, 10\}$	B	B	C
$\{1, 2, 3, 4, 6, 7, 8, 9, 10\}$	C	B	C

Minimization

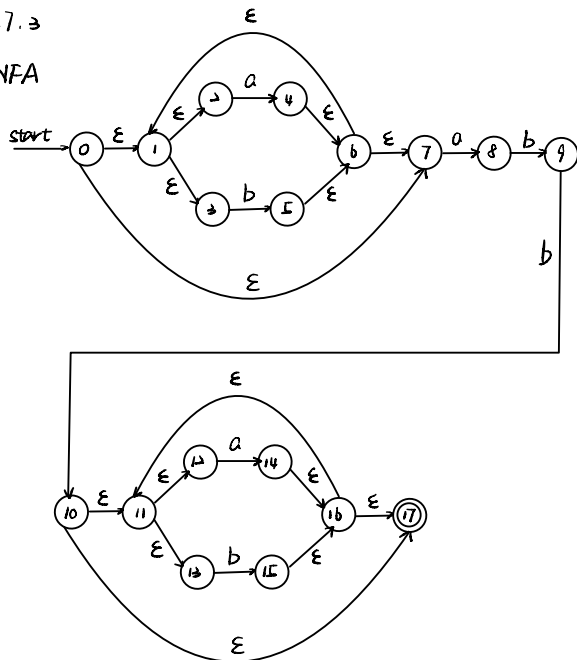
P_0	(A	B	C)
a	B	B	B
b	C	C	C
P_1	(A	B	C)

DFA



Ex 3.1.3

4) NFA



$$A = \epsilon\text{-closure}(0) = \{0, 1, 2, 3, 7\}$$

$$D_{\text{trans}}[A, a] = \{1, 2, 3, 4, 6, 7, 8\} =: B$$

$$D_{\text{trans}}[A, b] = \{1, 2, 3, 5, 6, 7\} =: C$$

$$D_{\text{trans}}[B, a] = B$$

$$D_{\text{trans}}[B, b] = \{1, 2, 3, 5, 6, 7, 9\} =: D$$

$$D_{\text{trans}}[C, a] = B$$

$$D_{\text{trans}}[C, b] = C$$

$$D_{\text{trans}}[D, a] = B$$

$$D_{\text{trans}}[D, b] = \{1, 2, 3, 5, 6, 7, 10, 11, 12, 13, 17\} =: E$$

$$D_{\text{trans}}[E, a] = \{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 16, 17\} =: F$$

$$D_{\text{trans}}[E, b] = \{1, 2, 3, 5, 6, 7, 11, 12, 13, 15, 16, 17\} =: G$$

$$D_{\text{trans}}[F, a] = F$$

$$D_{\text{trans}}[F, b] = \{1, 2, 3, 5, 6, 7, 9, 11, 12, 13, 15, 16, 17\} =: H$$

$$D_{\text{trans}}[G, a] = F$$

$$D_{\text{trans}}[G, b] = G$$

$$D_{\text{trans}}[H, a] = F$$

$$D_{\text{trans}}[H, b] = \{1, 2, 3, 5, 6, 7, 10, 11, 12, 13, 15, 16, 17\} =: I$$

$$D_{\text{trans}}[I, a] = F$$

$$D_{\text{trans}}[I, b] = G$$

Minimization

	(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)
a	B	B	B	B	F	F	F	F	F
b	C	D	C	E	G	H	G	I	G

	(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)
a	B	B	B	B	F	F	F	F	F
b	C	D	C	E	G	H	G	I	G

	(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)
--	-----	-----	-----	-----	-----	-----	-----	-----	-----

